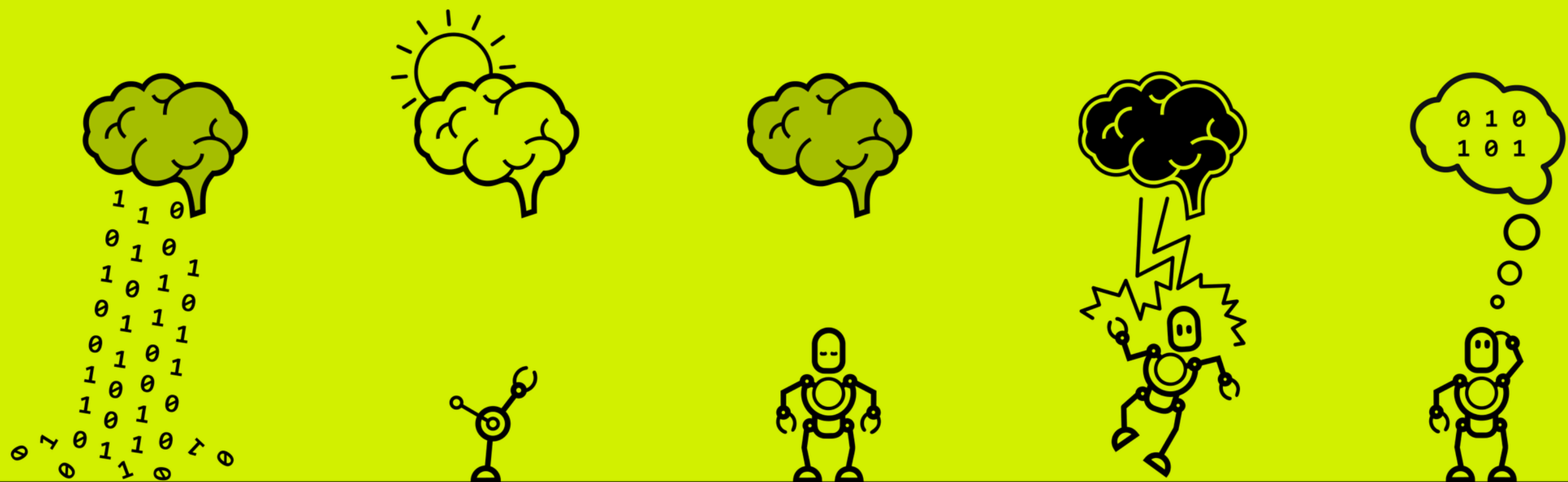




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THE BRAIN-COMPUTER INTERFACE
DESIGNERS HACKATHON

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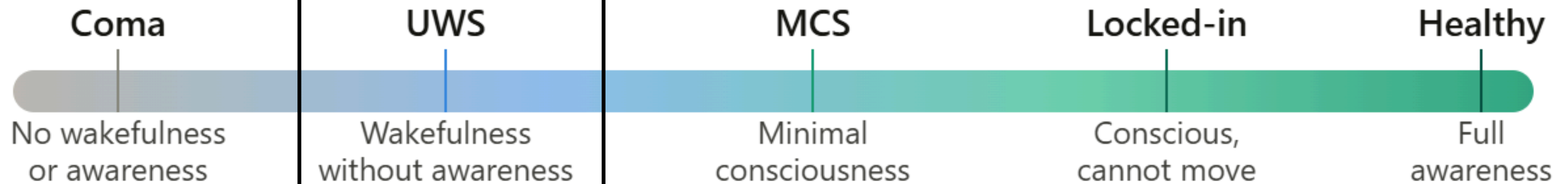
SILENCE IS NOT ABSENCE: EEG-BASED consciousness ASSESSMENT in UWS patients



Zidane Fatuna
Timoleon Fourfaro
Johana Dominguez
Neha

WHAT IS UWS?

- UWS: wakefulness without behavioural awareness.
- Caused by severe brain injury: trauma, stroke, oxygen deprivation.

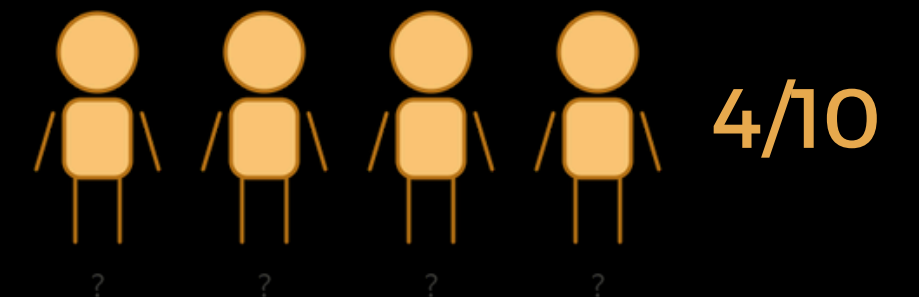


our patients

Correctly diagnosed as unresponsive

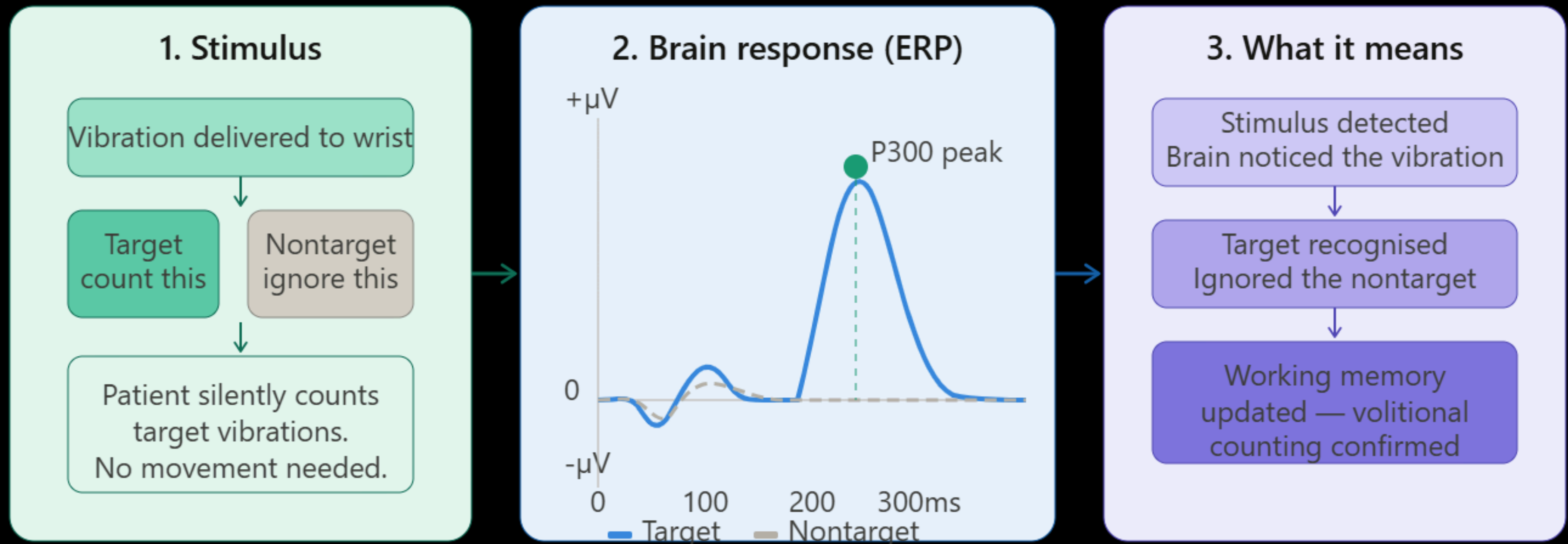


May have hidden awareness / misdiagnosed



THE P300

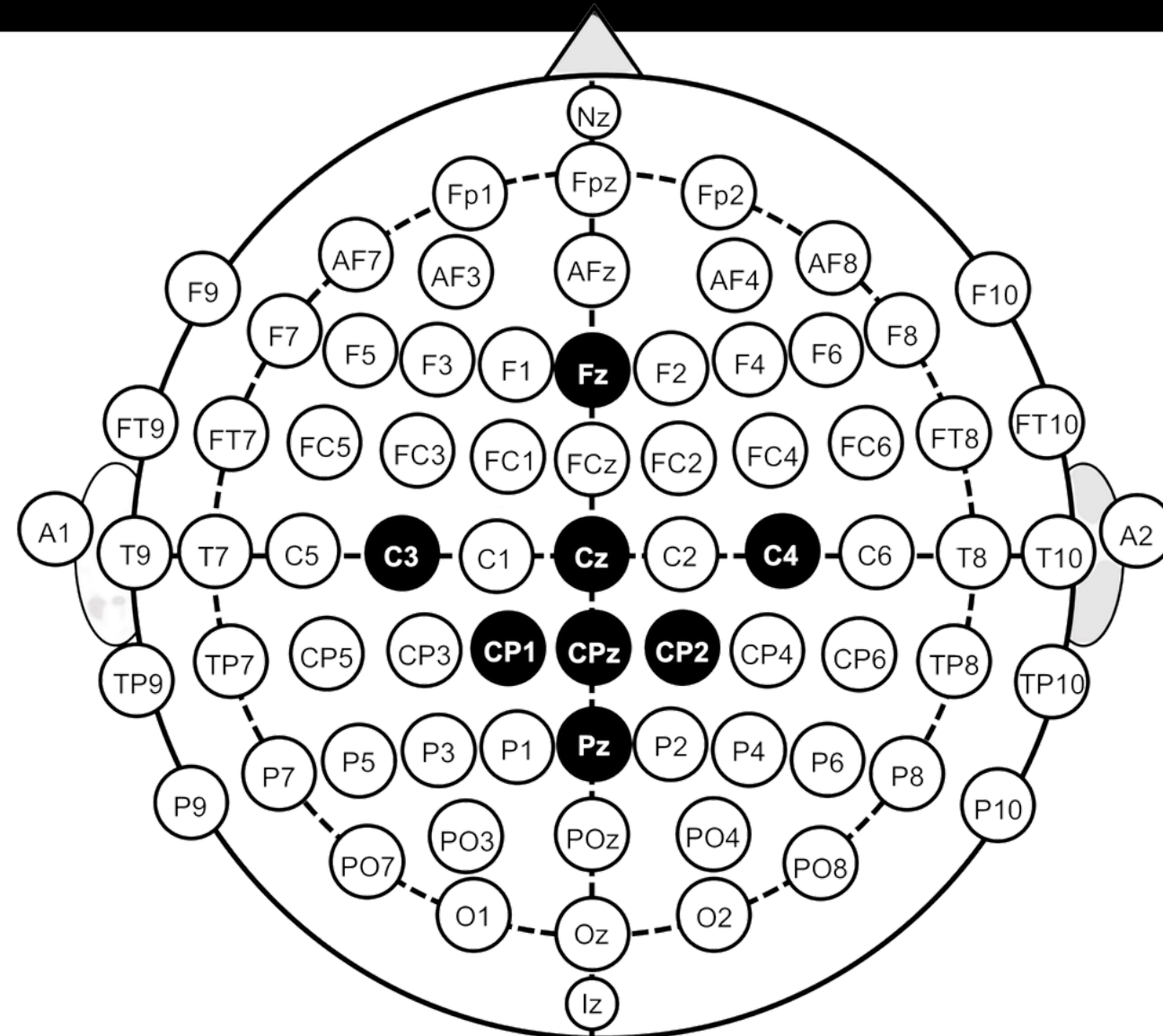
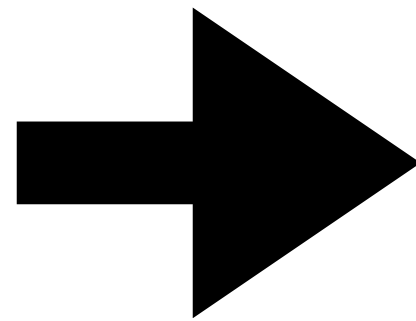
The P300 is a positive ERP component peaking at 300–500 ms post-stimulus over central and parietal scalp (Pz, CPz, Cz)



RECORDING SETUP & DATASET

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- 8-channel EEG targeting P300-sensitive central/parietal regions
- Oddball design (1 target : 1 non-target : 6 distractors)
- 480 stimuli/run, 4 sessions per patient (8 files total)



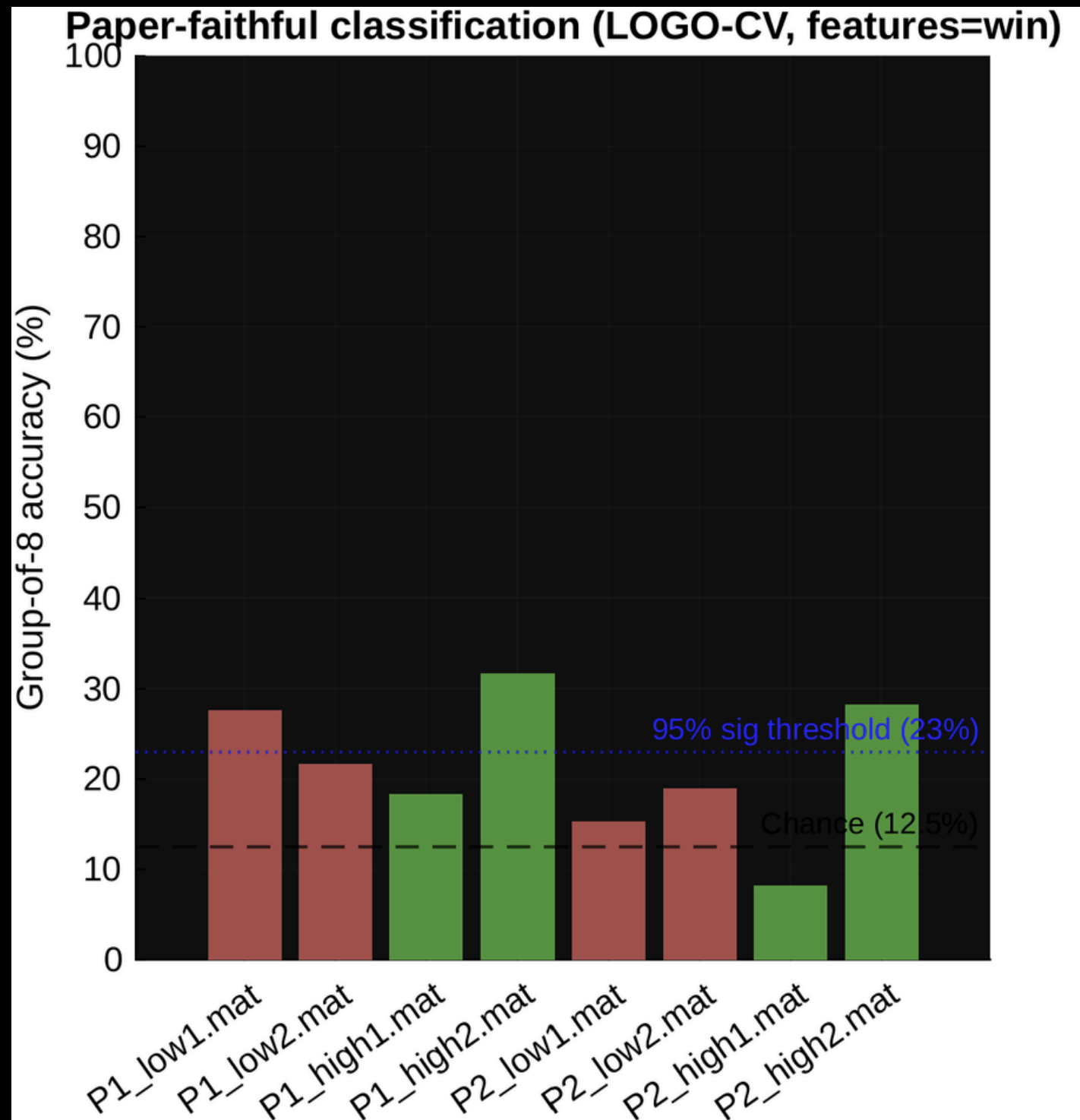
← **frontocentral**

← **frontoparietal**

← **parietal**

Analysis

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3 Observations

- 3 of 8 files clear the significance threshold
- The pattern does NOT track the high/low labels — significant files come from both conditions and both patients
- P2_high1 scored below chance (8.2%) despite being labeled as a >95% accuracy run

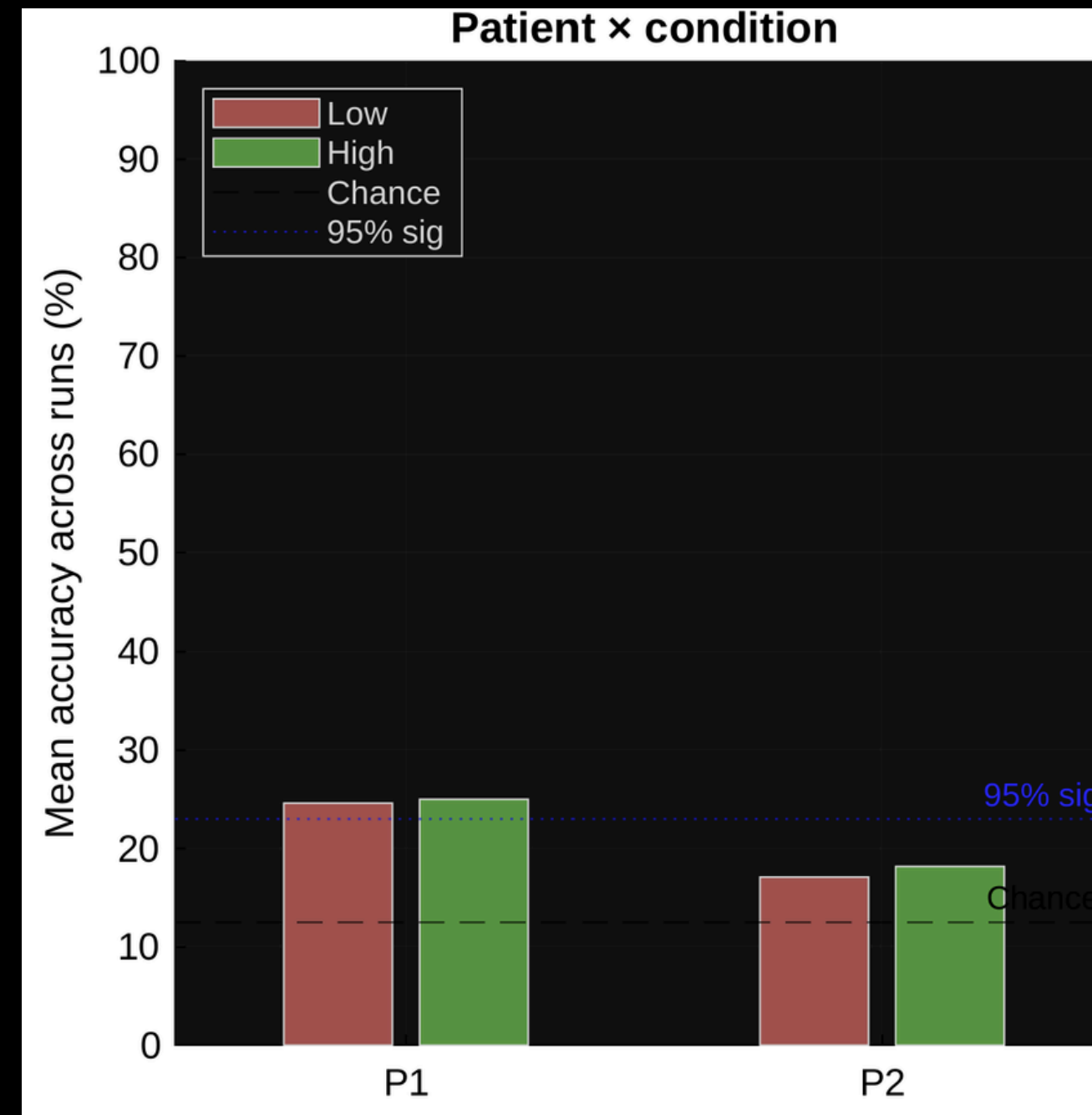
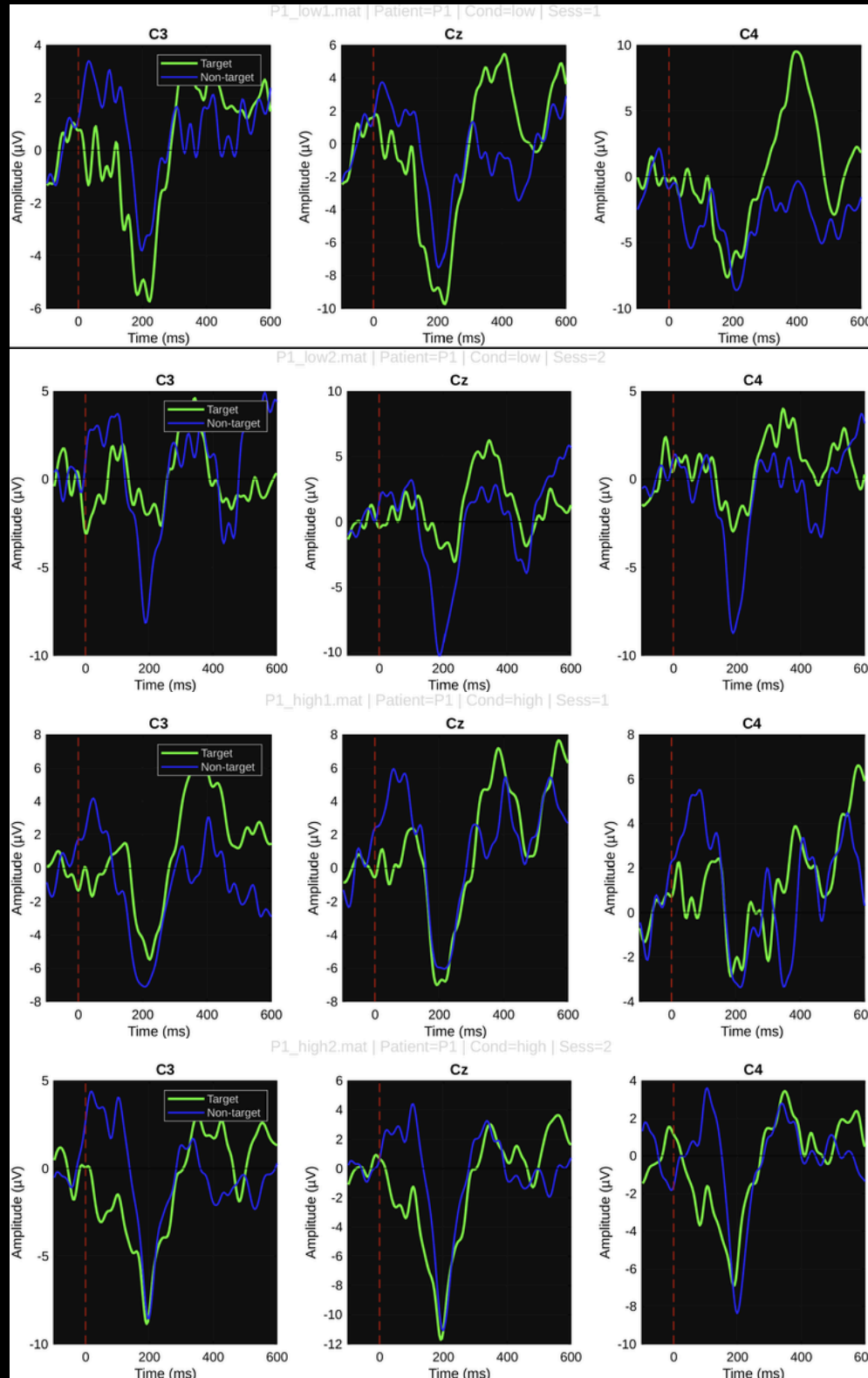
What does it mean?

- The dataset's labels were assigned by one classifier at one point in time
- Our faithful re-implementation of the published pipeline doesn't recover them
- The labels and our reanalysis disagree

Analysis

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Patient 1: The signal is there, labels don't predict it



3 Findings

- 2 of 4 runs cleared significance
- The two that cleared are split across labels — one "low," one "high"
- The original "high" run (high1) actually performed worse than one "low" run

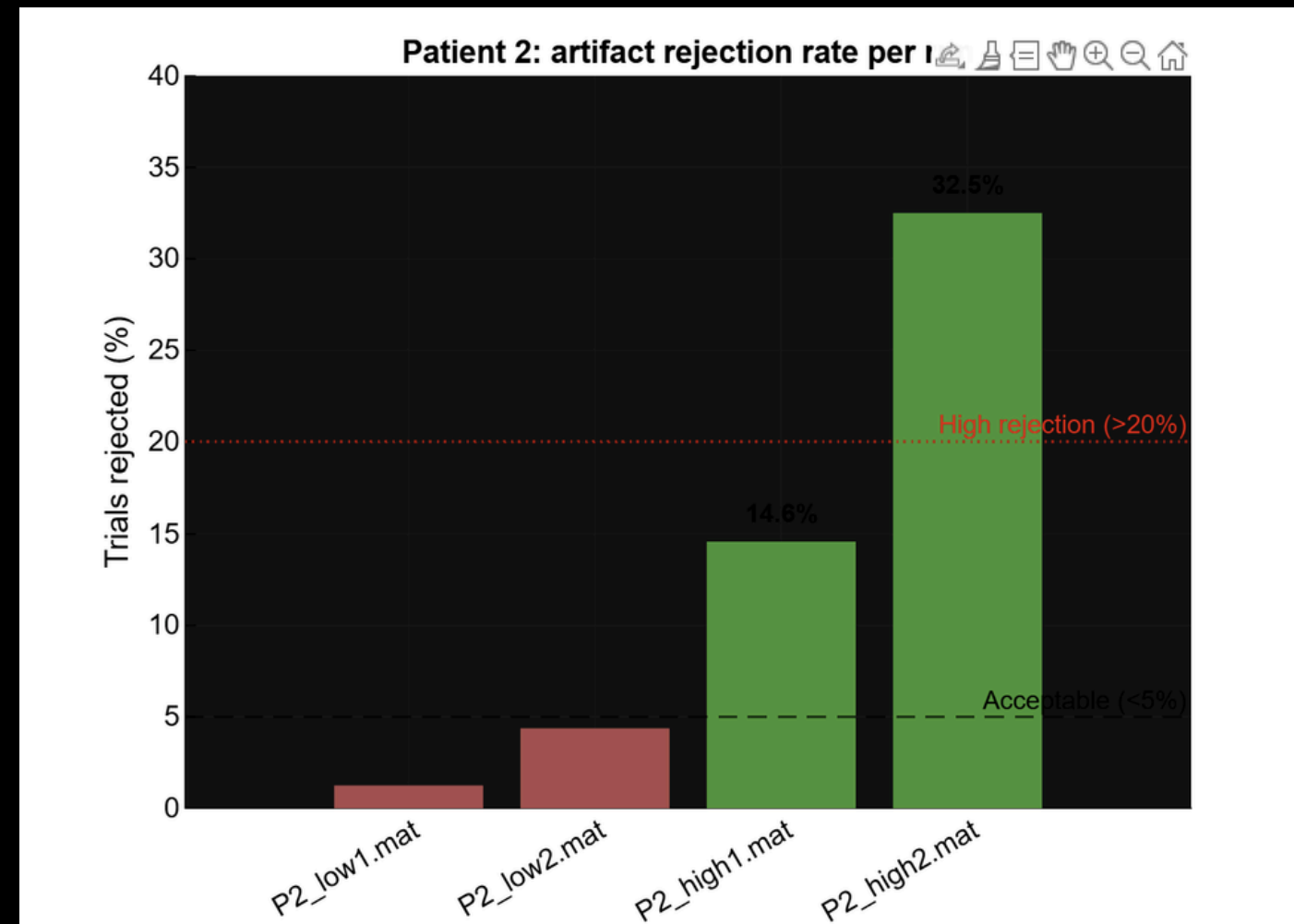
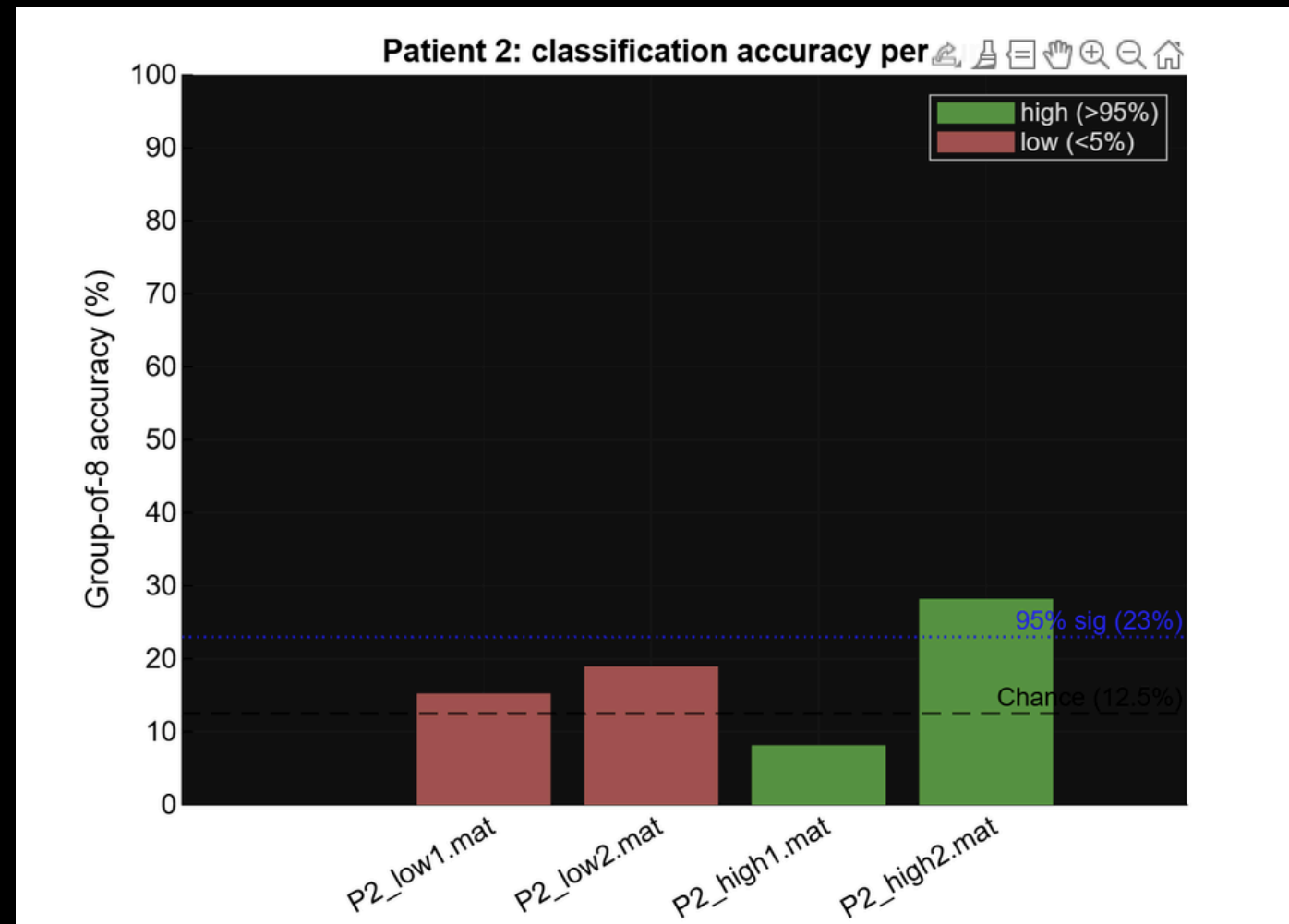
What does it mean?

- P1's brain shows discriminable target/non-target structure in some runs but not others
- The original labels don't predict which runs would replicate
- This is consistent with documented day-to-day fluctuation in UWS responsiveness — patients drift in and out of engagement across sessions

Analysis

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Patient 2: classifier success, signal quality issues



3 Findings

- Only 1 of 4 runs cleared significance
- One "high" run scored below chance (P2_high1 at 8.2%)
- The runs labeled as the most accurate are also the noisiest

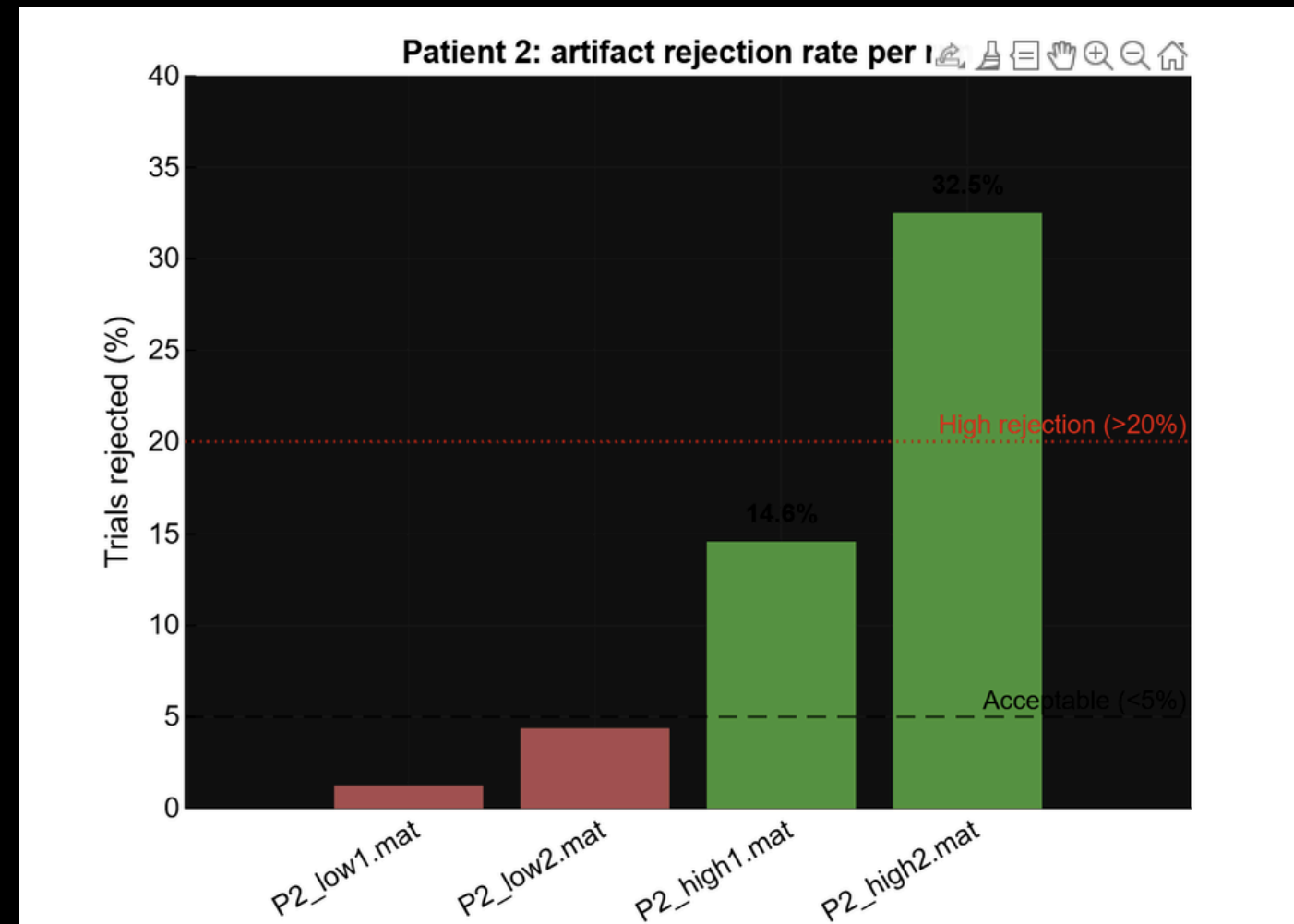
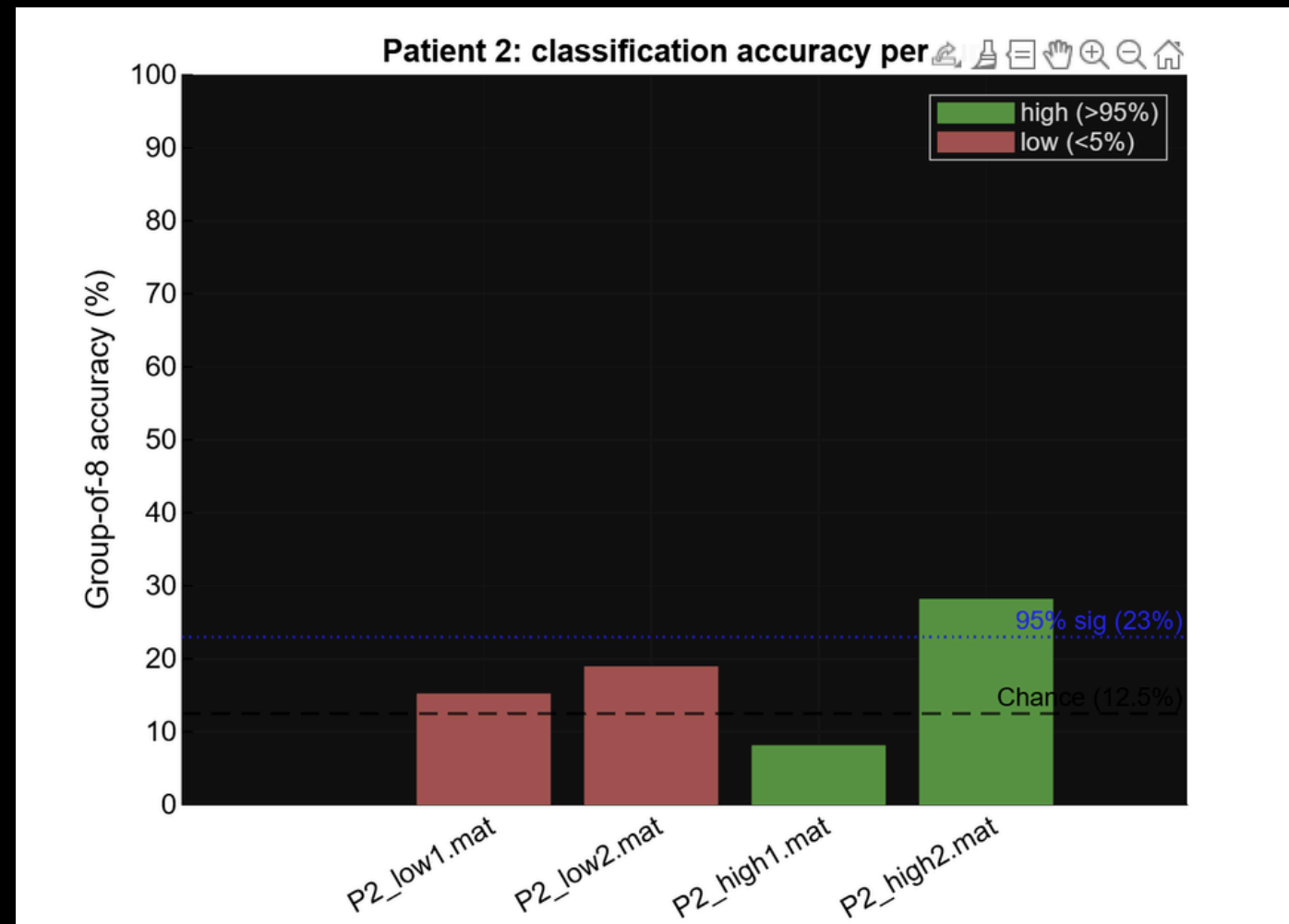
What does it mean?

- **Possibility 1:** real engagement was present, but mixed with movement artifacts
- **Possibility 2:** the original classifier was fitting noise patterns that don't survive reanalysis
- The data alone can't distinguish these — both have clinical implications

Analysis

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Patient 2: classifier success, signal quality issues



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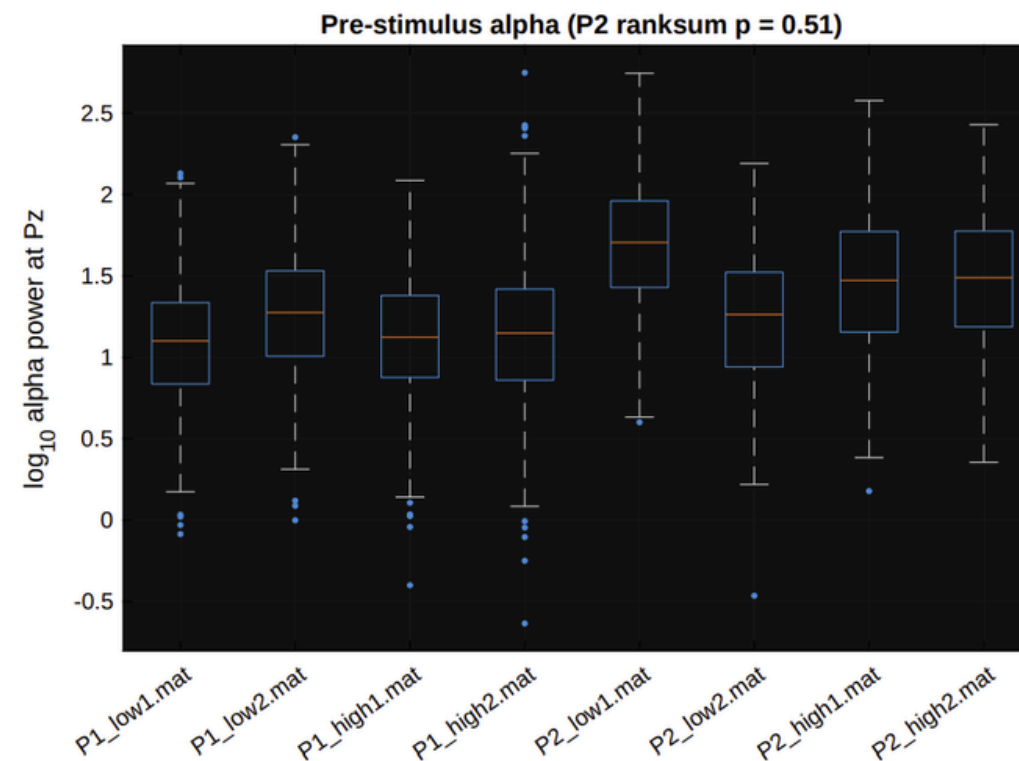
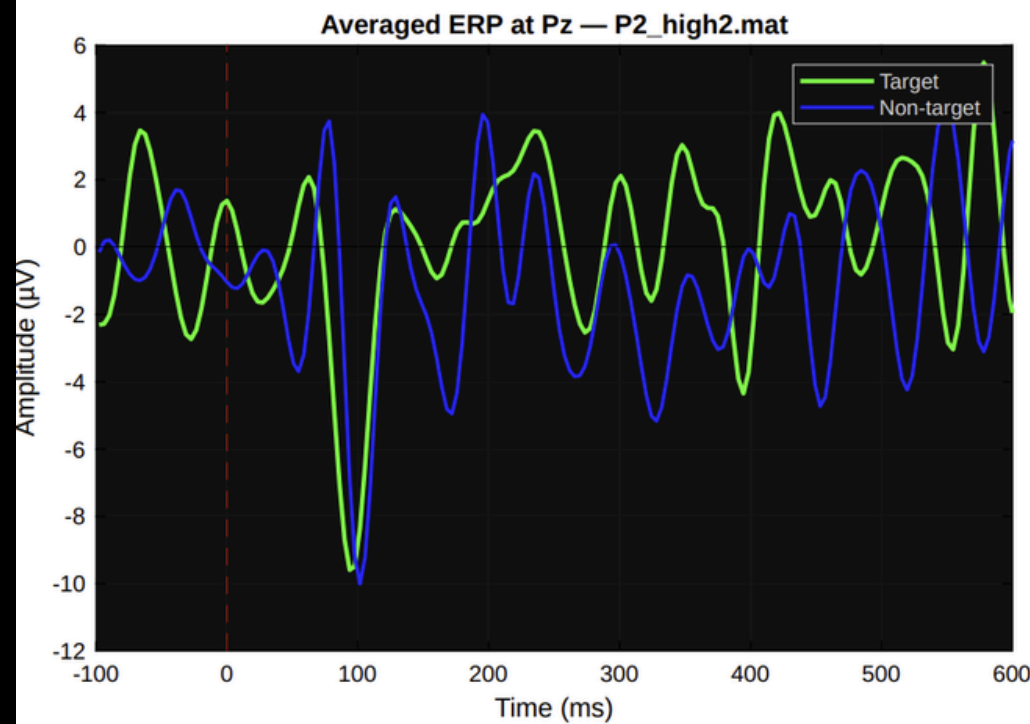
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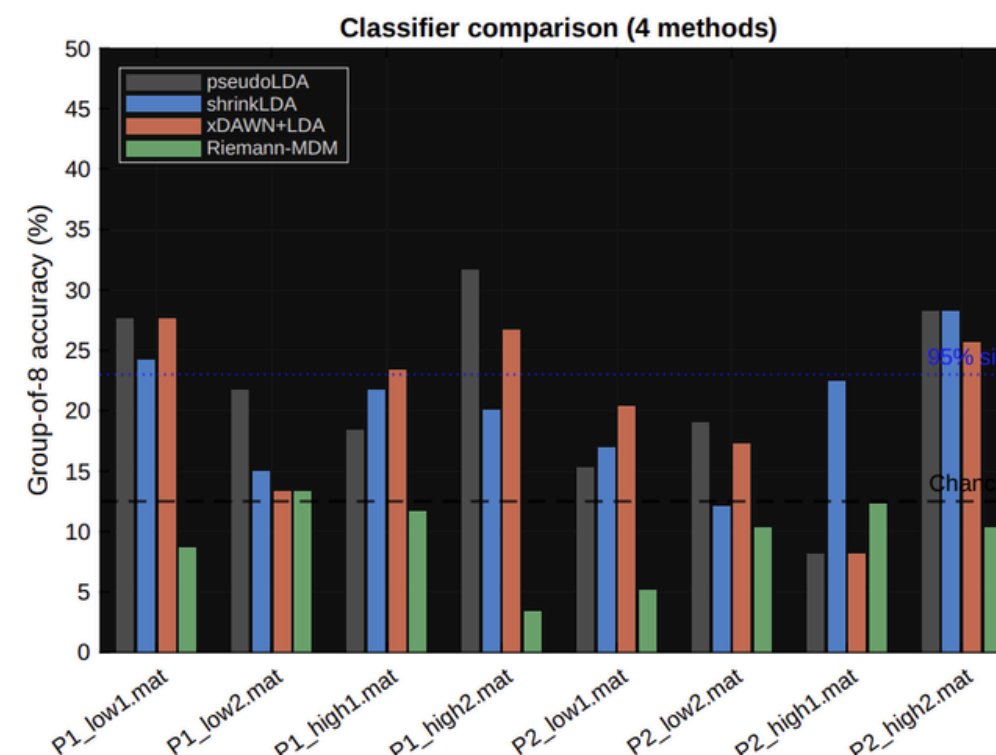
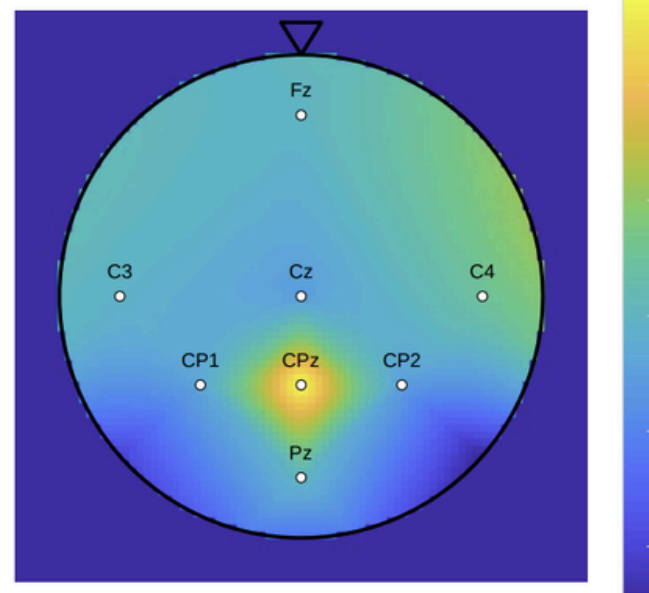
Analysis

What we looked for and didn't find

Null results: standard biomarkers do not differentiate conditions



Activation topography (P2_high2, 200–500 ms)



What we tested

- Averaged ERPs at parietal electrodes for canonical P300 morphology
- Pre-stimulus alpha power as a vigilance/engagement marker
- LDA activation patterns (Haufe-corrected) for the spatial location of discriminative information
- Three classifier alternatives — shrinkage LDA, xDAWN, Riemannian MDM — to test whether algorithm choice was the bottleneck

What we found

- No clean P300 in averaged ERPs, even in files that cleared classifier significance
- No significant alpha difference between high and low conditions (P2 ranksum $p = 0.51$)
- No canonical centro-parietal hotspot in classifier activation topographies
- No alternative classifier substantially exceeded the LDA baseline

Take Away

Single classifier accuracy can be fragile.

- Our replication didn't recover the dataset's high/low labels. Three of eight files cleared significance, scattered across both labels and both patients.

Algorithmic sophistication doesn't fix signal-level limits.

- Four classifier variants, including spatial filtering and Riemannian methods, gave essentially the same picture. The bottleneck is signal, not method.

Clinical use requires multi-session, multi-method, QC-aware assessment.

- Combining accuracy with signal-quality reporting, ERP inspection, and repeated sessions is the minimum bar for defensible interpretation.

References

- Giacino JT, Fins JJ, Laureys S, Schiff ND. (2014). Disorders of consciousness after acquired brain injury: the state of the science. *Nature Reviews Neurology*, 10:99–114.
- Spataro R, Heilinger A, Allison B, et al. (2018). Preserved somatosensory discrimination predicts consciousness recovery in unresponsive wakefulness syndrome. *Clinical Neurophysiology*, 129:1130–1136.
- Guger C, Spataro R, Pellas F, et al. (2018). Assessing command-following and communication with vibro-tactile P300 brain-computer interface tools in patients with unresponsive wakefulness syndrome. *Frontiers in Neuroscience*, 12:423.
- Wannez S, Heine L, Thonnard M, et al. (2017). The repetition of behavioral assessments in diagnosis of disorders of consciousness. *Annals of Neurology*, 81:883–889.

Questions?

GROUP PICTURE
PICS OF YOU WORKING...

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